

1. Executive Summary & Platform Overview

Samadhan.Connect is a state-wide collaborative platform engineered to crowdsource, triage, fund, and solve real-world civic challenges across Jharkhand. The platform connects five core user roles: **Citizens** (reporting issues), **Students & Innovators** (submitting technical solutions), **Universities & Faculty** (mentoring projects), **Industries & CSR Partners** (funding initiatives), and **Nodal Officers** (verifying and implementing solutions at the municipal/district level).

To deliver seamless performance, high security, and real-time civic intelligence, the platform combines a modern **React 19 Single Page Application (SPA)** frontend with a **Supabase PostgreSQL** relational database, **Vercel Serverless / Node.js API Gateway**, and an advanced **Google Gemini AI Assistant** restricted to 25 strict civic domains.

2. Complete Technology Stack

Layer	Technologies Used	Key Purpose & Architectural Role
Frontend Framework	React 19, Vite 8, React Router v7	Ultra-fast SPA rendering, component modularity, client-side routing, optimistic state updates.
Styling & Design System	Tailwind CSS v3, PostCSS, Lucide Icons	Modern glassmorphism, responsive grid layouts, cohesive typography, high-contrast accessible UI.
Authentication & Authorization	Supabase Auth, OAuth 2.0 (Google, GitHub)	Passwordless secure OAuth login, JWT session management, first-time role onboarding modal, RBAC.
Database & Persistence	Supabase (PostgreSQL 15), Row Level Security (RLS)	Relational schema (profiles, problems, solutions, projects, messages), auto-triggers, real-time sync.
Backend & API Gateway	Vercel Serverless Functions (/api/chat.js), Vite Dev Middleware, Node.js Express	Zero-exposure environment secrets handling, request validation, CORS, rate limiting, and AI routing.
Artificial Intelligence Engine	Google Gemini API (@google/genai SDK, gemini-2.5-pro, gemini-2.5-flash)	25 strict civic domain guidance, out-of-domain rejection, step-by-step problem-solving advice.
Micro-Interactions	Canvas Confetti, Real-time status badges	Delightful user feedback for milestone achievements, problem approvals, and solution submissions.

3. How the Backend Works (Simple Plain Language)

Think of the **Samadhan.Connect Backend** as an invisible, highly secure coordinating bridge between the citizen's browser, the central database, and the Google Gemini AI brain. Here is how it functions in 4 simple concepts:

A. Secure Gatekeeper (Serverless API Gateway)

When a user performs an action—such as asking a question to **Samadhan AI** or submitting a project update—the request does *not* talk directly to third-party services. Instead, it sends an encrypted HTTP request to `/api/chat`. The backend acts as a security guard: it inspects the request, confirms the user is not sending malicious data, injects secret server-only credentials (like `GEMINI_API_KEY`), and protects confidential keys from ever reaching the browser.

B. Smart Dual-Engine Router (Live Gemini + Offline Knowledge Base)

The backend is built with **zero-downtime resilience**. When a chat message arrives, it first checks if live Google Gemini models (like `gemini-2.5-pro` or `gemini-2.5-flash`) are accessible. If the API key is active, it runs Gemini with the strict 25-domain civic prompt. If the API is offline or quota-limited, the backend automatically switches to its **Intelligent Civic Knowledge Engine**, instantly returning accurate, verified answers for Jharkhand schemes, electricity complaints, or farming subsidies without failing.

C. Relational Database & Real-Time Sync (Supabase PostgreSQL)

All data (user profiles, reported civic issues, innovation proposals, funding pledges, and peer chat messages) is stored inside a PostgreSQL database. Database Triggers automatically create profile rows when users log in via Google/GitHub OAuth, while Row Level Security (RLS) ensures citizens only edit their own problems and Nodal Officers only verify issues in their jurisdictions.

D. Role-Based Access Control (RBAC)

The backend guarantees strict authorization. The super-admin (`microsoft1gab@gmail.com`) has full privileges to promote any user to an Admin, while role filters dynamically tailor the dashboard views for Students, Universities, Industries, and Government Officers.

4. End-to-End System Architecture Diagram

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CLIENT TIER (React 19 / Vite SPA) | | [Citizen Dashboard] [Student Lab] [University Desk] [Industry CSR
Portal] [Nodal Admin Hub] | | • OAuth Sign-In (Google/GitHub) • Problem Reporting Form • Samadhan AI Chatbot
UI | +-----+ ■
■ REST / Supabase JS API POST /api/chat ■ ■ ▼ ▼ +-----+
+-----+ | DATABASE & AUTH TIER (Supabase) | | BACKEND / API GATEWAY TIER
| | • PostgreSQL 15 Relational DB | | • Vercel Serverless (api/chat.js) | | • Supabase Auth (JWT & OAuth
Handlers) | | • Vite Dev Server Custom Middleware | | • Row Level Security (RLS) Policies | | • Server Config
(chatbotPrompt.js) | | • Database Triggers & Auto-Profile Sync | | • Secure process.env (GEMINI_API_KEY) | | •
Tables: profiles, problems, solutions, projects | | • Multi-Model Retry & Timeout Controller|
+-----+ +-----+ +-----+ ■
@google/genai SDK Call ■ ▼ +-----+ | EXTERNAL AI CLOUD TIER | | • Google
Gemini 2.5 Pro / Flash Engine | | • Strict 25 Civic Domain Guardrail | | • Step-by-Step Problem Solving Logic
| +-----+
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5. Authentication & Role Onboarding Lifecycle

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[ User clicks 'Sign in with Google' or 'Sign in with GitHub' ] ■ ▼ [ Supabase OAuth handles 3rd-Party Authentication & returns JWT ] ■ ▼ [ PostgreSQL Auth Trigger checks if profile exists in 'profiles' table ]
■■■ Existing User ■■> [ Load Profile & Redirect to Role Workspace ] ■■■ New User ■■■■■■■■■■> [ Insert Basic Profile & Flag 'needs_role_selection' ] ■ ▼ [ Frontend detects missing role and triggers 'RoleSelectionModal' ]
[ User selects 1 of 5 Roles: Citizen | Student | University | Industry | Nodal Officer ] ■ ▼ [ DataService updates profile in Supabase DB with selected Role ] ■ ▼ [ Special Condition Check: Is email == 'microsoft1gab@gmail.com'? ] ■■■ YES ■■> [ Grant SUPER_ADMIN role with full /admin panel privileges ] ■■■ NO ■■> [ Grant Standard Verified Role Dashboard Access ]
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6. Samadhan AI Chatbot Request-Response Lifecycle

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[ User types message or clicks Suggested Domain Chip (e.g. '■ Agriculture') ] ■ ▼ [ Frontend Appends Message to State + Truncates History to last 10 messages ] ■ ▼ [ POST /api/chat payload: { message: "...", history: [...] } ] ■ ▼ [ Backend Gateway reads process.env.GEMINI_API_KEY ] ■■■ API Key Present ■■> [ Initialize @google/genai with SAMADHAN_SYSTEM_PROMPT ] ■ ■ ■ ▼ ■ [ Send to Gemini 2.5 Pro / Flash Engine ] ■ ■■■ Success ■■> [ Return JSON reply & status: success ] ■ ■■■ Rate Limit/Error ■■> [ Fallback to Local Knowledge ] ■ ■■■ No API Key ■■■■■■■■■■> [ Run Local Knowledge Engine for 25 Civic Domains ] ■■■ Is In-Domain? ■■■■■> [ Return Structured Civic Steps ] ■■■ Is Out-of-Domain? ■■> [ Return Official Refusal Text ] ■ ▼ [ Frontend receives JSON response & renders via MarkdownMessage component ] [ Formats Bold text, Bullet lists, Numbered steps, and Verification disclaimers ]
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7. Civic Problem-Solving & Collaboration Pipeline

1. REPORT PHASE

- Citizen logs in and submits a localized problem (e.g. "Water contamination in Palamu ward 4").
- Uploads photos, location coordinates, category, and urgency level.
- System assigns unique Tracking ID & notifies District Nodal Officer.

2. TRIAGE & VALIDATION PHASE

- Nodal Officer verifies authenticity, inspects feasibility, and approves as an "Active Challenge".
- Problem is published on the Public Challenge Feed for academic & startup incubation.

3. SOLUTION & PROPOSAL PHASE

- Students & University Innovation Cells form multidisciplinary teams.
- Submit technical proposals (design blueprints, budget estimate, timeline, prototype demo).
- Faculty Mentors review and endorse the academic rigour of the solution.

4. CSR FUNDING & MATCHING PHASE

- Industry & CSR Partners browse validated project proposals.
- Pledge corporate grants, CSR funding, or technical mentorship.

5. DEPLOYMENT & VERIFICATION PHASE

- Solution is implemented on the ground.
- Nodal Officer confirms operational completion; Citizen receives resolution SMS/notification.

8. Security, Data Integrity & Deployment Guarantees

Security Dimension	Implementation in Samadhan.Connect
API Key Isolation	GEMINI_API_KEY is restricted to server-side Node.js / Vercel Serverless execution. It is never bundled into client JS or exposed in browser storage.
Row Level Security (RLS)	PostgreSQL database tables enforce RLS policies preventing unauthorized access across user tenants and role boundaries.
Out-of-Domain Guardrails	System prompt strictly confines AI answers to the 25 authorized civic domains, rejecting entertainment, gaming, jokes, and irrelevant queries.
Factual AI Disclaimers	AI assistant explicitly adds disclaimers for healthcare, legal, and banking advice, directing citizens to official portals like jharsewa.jharkhand.gov.in.
Super Admin Safeguards	Elevated privileges to promote users to Admin are restricted strictly to verified master administrative accounts (microsoft1gab@gmail.com).

Document Summary: This technical specification details the complete frontend, backend, database, and AI architecture powering Samadhan.Connect. Designed for scalability, high availability, and rapid citizen problem resolution.